SUGGESTED SOLUTIONS TO 2023 H2 PRELIM PAPER 1

Q1.
Check for homogeneity of equations. An inhomogeneous equation is an incorrect equation.

Let base units be represented by BU.

A: LHS: BU = kg m^2 s^{-2}
RHS: BU = (kg m $s^{-1)}$(m s^{-1}) = kg m^2 s^{-2}
LHS = RHS $\Rightarrow$ homogeneous equation (wrong answer)

B: LHS : BU = (kg m^2 s^{-2})^2 = kg^2 m^4 s^{-4}
RHS: BU of p^2c^2 = (kg m $s^{-1)2}$ (m s^{-1})^2 = kg^2 m^4 s^{-4}
BU of m^2c^4 = kg^2 (m s^{-1})^4 = kg^2 m^4 s^{-4}
All terms on both sides of the equal sign have the same units $\Rightarrow$ equation is homogeneous (wrong answer)

C: LHS: BU = kg m^2 s^{-2}
RHS: BU =(kg m $s^{-1)\,2}$(m s^{-1}) = kg m^3 s^{-3} $\Rightarrow$ inhomogeneous equation (correct answer)

D: LHS: BU = kg m^2 s^{-2}
RHS: BU = $\dfrac{\left(\text{kg m s}^{-1}\right)^2}{\text{kg}} = \text{kg m}^2\ \text{s}^{-2}$ $\Rightarrow$homogeneous equation (wrong answer)

Ans: C

Q2

$$B = \frac{AD}{6C^2} \Rightarrow \frac{\Delta B}{B} = \frac{\Delta A}{A} + \frac{\Delta D}{D} + 2\frac{\Delta C}{C}$$

$$\frac{\Delta B}{B} = \frac{0.1}{2.0} + \frac{0.01}{3.45} + 2\frac{0.05}{4.10} = 7.7x10^{-2} = 7.7\%$$

Ans: C

Q3
Area under *v*-*t* graph is the displacement.

For graph P, the area is always positive, meaning that there is positive displacement.
For graph Q, the area during the negative velocities and the area during positive velocities are equal. Hence the total displacement is zero.
Displacements for graphs P and Q are thus different.
The magnitudes of the total area under the graphs P and Q are *different*. Thus the total distances travelled for P and Q is *different*.

Ans: A

Q4
Here the vertical forces are gravitational and magnetic forces. The horizontal forces are the electric force and air resistance.

The electrically charged oil drop is moving horizontally at constant velocity. The oil drop is in dynamic equilibrium and the resultant force is zero in all directions.

Vertically, the gravitational force is acting downwards. Therefore, the magnetic force must be acting upwards to ensure equilibrium.

The electric force cannot act vertically in this case because the electric field is horizontal, and air resistance should be acting in the direction opposite to the motion, which is horizontal in this case.

Ans: D

Q5

When floating, net force on steel = 0. Therefore, Upthrust – weight = 0

$V_{submerged} \times \rho_{mercury} \times g = V_{steel} \times \rho_{steel} \times g$

Hence, fraction of steel submerged

$= V_{submerged} / V_{steel} = \rho_{steel} / \rho_{mercury}$ = 8050 / 13600

= 0.59

Ans: C

Q6

For the mass of 500 g to move downwards and the mass of 200 g to move upwards, the friction F in the pulley must act opposite to the motion of the masses.

For the masses to move at constant speed, net force = 0. Considering the pulley and masses as a whole, $Mg - F - mg = 0$ or $F = (M - m)g$

F = (0.500 – 0.200) x 9.81 = 2.94 N

Speed of rope = $v = r\,\omega$ = 0.035 x 2.4 = 0.084 m s^{-1}

Hence, rate of work done = $P = F\,v$ = 2.94 x 0.084 = 0.25 W

Since the force exerted by the motor on the rope is opposite in direction to the velocity of the rope, the rate of work done = - 0.25 W

Ans: B

Q7.

From graph, $F = kx$ or $4.5 = k\,(0.200 - 0.050) \Rightarrow k = 30.0$ N m^{-1}

By the Law of Energy Conservation, Loss in GPE of load = Gain in elastic PE of spring

$mgx = (1/2)kx^2$ or $0.150 \times 9.81\,(x) = (1/2)\,(30.0)\,x^2$

$x\,(15.0\,x - 1.47) = 0$ or $x = 0.0981$ m = 9.81 cm

So, maximum length attained = 5.0 + 9.81 = 14.8 cm

Ans: C

Q8

The resultant gravitational field on the 1 kg mass is zero. Therefore, $\dfrac{GM_1M_2}{x^2} = \dfrac{GM_2M_3}{(15-x)^2}$

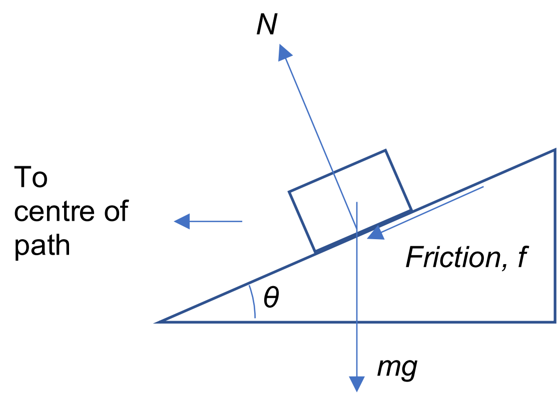
$\frac{M_3}{M_1} = \frac{(15-x)^2}{x^2} \Rightarrow \sqrt{\frac{16}{9}} = \frac{15-x}{x}$

$\frac{4}{3} = \frac{15-x}{x}$ or $4x = 45 - 3x$

Hence x = 6.4 m

Ans: C

Q9



The centripetal force for circular motion is constituted by the sum of the component forces of *N* and *f* towards the centre of the circular path. Hence

$$N\sin\theta + f\cos\theta = \frac{mv^2}{r}$$

Ans: B

Q10

A faster tapping of the foot in the water corresponds to a higher frequency of the waves produced. The velocity of the waves depends upon the medium which is unchanged, and hence remains constant.

Ans: A

Q11

The frequency of oscillation is determined by the driver.

$f = \frac{1}{T} = \frac{1}{5}$ = 0.20 Hz

Ans: A

Q12

Since no energy is lost to the surroundings (insulated container),

Initial total KE of molecules in both gases = Final total KE of molecules of both gases (at common temperature)

(3/2) $\mu_1 R T_1$ + (3/2) $\mu_2 R T_2$ = (3/2) $(\mu_1 + \mu_2) R T$ where μ refers to number of moles of gas.

Hence $\mu_1 T_1 + \mu_2 T_2 = (\mu_1 + \mu_2) T$

$3.0 \times 400 + 5.0 \times 200 = (3.0 + 5.0) T$ or $T = 275$ K

Ans: B

Q13

Diffraction rating formula: $n\lambda = d\sin\theta_1$

Consider $n = 1$. Then $5.30 \times 10^{-7} = d\sin 15.4^\circ$ or $d = 1.996 \times 10^{-6}$ m

For n = 2, we have $2\lambda = d\sin\theta_2$

$2 \times 5.30 \times 10^{-7} = 1.996 \times 10^{-6} \sin\theta_2$

$\theta_2 = 32.1^\circ$.

Hence, $\theta_2 - \theta_1 = 16.7^\circ$

Ans: C

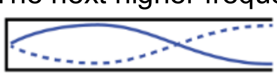
Q14

Let L be the length of the bugle, v be the speed of sound in air, λ be the wavelength.

92 Hz corresponds to that shown below:

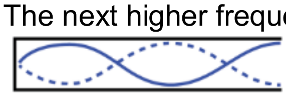
$4L = \lambda_1 = v/92$, $v = 368L$

The next higher frequency corresponds to that shown below:



$3\lambda_2/4 = L$ or $3(v/f_2) = 4L$ or $3(368L/f_2) = 4L$, $f_2 = 276$ Hz

The next higher frequency corresponds to that shown below:



$5\lambda_3/4 = L$, $5(v/f_3) = 4L$, $5(368L/f_3) = 4L$, $f_3 = 460$ Hz

Ans: C

Q15

The p.d. across the $4\,\Omega$ resistor, $V = IR = 1.5 \times 4 = 6.0\,\text{V}$

Hence the current in the $13\,\Omega$ resistor, $I = \dfrac{V}{R} = \dfrac{6.0}{13} = 0.4615\,\text{A}$

The total current flowing in the internal resistor = $0.4615 + 1.5 = 1.9615\,\text{A}$

Using $P = IV$, we get $V = \dfrac{2.1}{1.9615} = 1.0706 = 1.1$ V

Hence emf = total pd = 3.0 + 6.0 + 1.1 = 10.1 V

Ans: D

Q16

The resistance should start at zero (s = 0), increase in magnitude and then end at zero (s = L).
Let resistance per unit length of the wire be k.
The resistance between X and Y will be that of two resistors in parallel.

Thus $R = \left(\dfrac{1}{R_1} + \dfrac{1}{R_2}\right)^{-1}$ = $\dfrac{R_1 R_2}{R_1 + R_2}$

or $R = \dfrac{(ks)[k(L-s)]}{ks + k(L-s)} = \dfrac{ks(L-s)}{L}$

The variation of R with s is therefore not a straight-line relationship.

Ans: B

Q17

Since the 10 Ω resistor and $30\,\Omega$ resistor are in parallel, they have the same p.d.

$P_{10\Omega} = \dfrac{V^2}{10}$, $P_{30\Omega} = \dfrac{V^2}{30}$

Hence $\quad P_{10\Omega} = 3 \times P_{30\Omega}$

$\therefore \dfrac{P_{10\Omega}}{P_{total}} = \dfrac{3 \times P_{30\Omega}}{3 \times P_{30\Omega} + P_{30\Omega}} = \dfrac{3}{4}$ or $P_{10\Omega} = \dfrac{3}{4} \times P$ while $P_{30\Omega} = \dfrac{1}{4} P$.

Ans: B

Q18

Object A experiences a force F_A equal to Eq in the direction of the force. Starting from rest, its velocity at time t is $v_A = 0 + \dfrac{Eq}{m} t$. Hence its KE at time t is

$K_A = \dfrac{1}{2} m v_A^2 = \dfrac{1}{2} m \left(\dfrac{E^2 q^2 t^2}{m^2}\right) = \dfrac{E^2 q^2 t^2}{2m}$. In gaining this KE, it would have lost an equal amount of EPE.

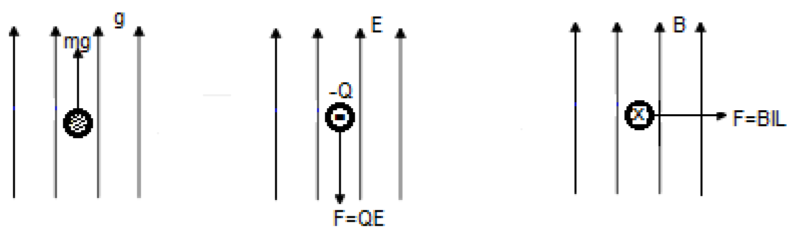
Object B experiences a force F_B equal to $E(2q)$ in the direction opposite to that of the force. Starting from rest, its velocity at time t is $v_B = 0 + \dfrac{E(2q)}{2m} t = \dfrac{Eqt}{m}$. Hence its KE at time t is

$K_B = \frac{1}{2}(2m)v_B^2 = \frac{1}{2}(2m)\left(\frac{E^2q^2t^2}{m^2}\right) = \frac{E^2q^2t^2}{m}$. In gaining this KE, it would have lost an equal amount of EPE.

Comparing the two scenarios, object B would have lost twice the EPE compared to A at an instant after some time from rest.

Ans: D

Q19



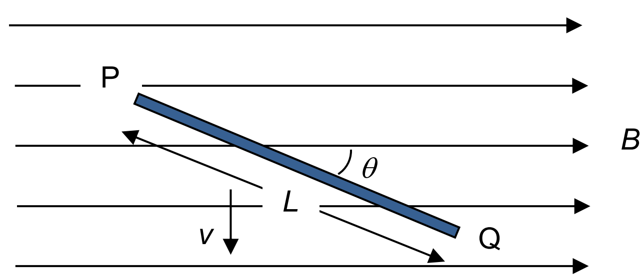
Ans: D

Q20

The currents in the two wires are opposite and hence the two wires repel. The magnitude of the forces acting on each wire is the same by Newton's third law.

Ans: C

Q21



Using Fleming's Right Hand Rule, the induced emf is across the width of the rod, and there is no emf between the ends of the rod.

Ans: A

Q22

Based on Lenz's law, the direction of the induced current in the inner loop should be in same direction of the current in the outer loop to oppose the decrease in B-field from the outer loop.

The induced current decreases over time such that when the current in the outer loop drops to zero, the induced current in the inner loop also drops to zero.

Ans: A

Q23

Note: A diode arranged in series with a resistor will give a half-wave rectified waveform with a rms current $I_{rms} = \dfrac{I_0}{2}$.

When the 1000 Ω resistor is conducting, peak current = 12/1000 = 0.012 A. Root mean square current is

$$I_1 = \frac{0.012}{2} = 0.0060 \text{ A (first half cycle)}$$

When the 2000 Ω resistor is conducting, peak current = 12/2000 = 0.0060 A. Root mean square current is

$$I_2 = \frac{0.0060}{2} = 0.0030 \text{ A (second half cycle)}$$

Total Average power $< P >= I_1^2 R_1 + I_2^2 R_2 = 0.006^2(1000) + 0.003^2(2000)$ = 0.054 W

Ans: B

Q24

$hf = eV_s + \Phi,$

$f_b > f_a$ since $V_{sb} > V_{sa}$

Since $i_a < i_b$, $I_a < I_b$

Ans: B

Q25

For potassium, $hc/\lambda = \Phi_{pot} + (1/2)mv_{max}^2$

$hc/(250\text{x}10^{-9}) = 2.24\text{x}1.6\text{x}10^{-19} + (1/2)(9.11\text{x}10^{-31})v_{max}^2$

$v_{max} = 9.80\text{x}10^5$ m s^{-1}

For zinc, $hc/\lambda = \Phi_{zn} + (1/2)mv_{max}^2$

$hc/(250\text{x}10^{-9}) = 4.33\text{x}1.6\text{x}10^{-19} + (1/2)(9.11\text{x}10^{-31})v_{max}^2$

$v_{max} = 4.75\text{x}10^5$ m s^{-1}

Ratio of speeds = $9.80\text{x}10^5/(4.75\text{x}10^5)$ = 2.1

Ans: C

Q26

An electron can give a part of its energy to excite the sodium atom. In this case, it can give up to a maximum of 5.68 $\times 10^{-19}$ J to the atom, thus exciting the atom to energy level n = 4. From there on, the atom can de-excite to the ground level in 6 ways, thus producing 6 corresponding emission spectral lines.

Ans: D

Q27

X-ray spectra are created by electrons impinging on a metal.

Loss in KE of electrons = Energy of X-rays

Maximum KE of electrons - 0 = Maximum energy of X-rays

Maximum KE of electrons = $E = \frac{hc}{\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{35 \times 10^{-12}} = 5.68 \times 10^{-15}$ J

Ans: C

Q28

Binding energy = mass defect x c^2

= (mass of nucleons – mass of nucleus) x c^2

= [(2 x 1.0087)+ (2 x 1.0073) – 4.0015] x 931 MeV = 28.4 MeV

Ans: A

Q29

Ans: C

Q30

$$A = \lambda N = \left(\frac{\ln 2}{t_{\frac{1}{2}}}\right)\left(\frac{M}{M_m} N_A\right) \text{ or}$$

$$M = \frac{A t_{\frac{1}{2}} M_m}{(\ln 2) N_A} = \frac{(3.7x10^{10})(138x24x60x60)(210)}{(\ln 2)(6.02x10^{23})} = 2.2x10^{-4} \text{ g}$$

= 0.22 mg

Ans: A

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